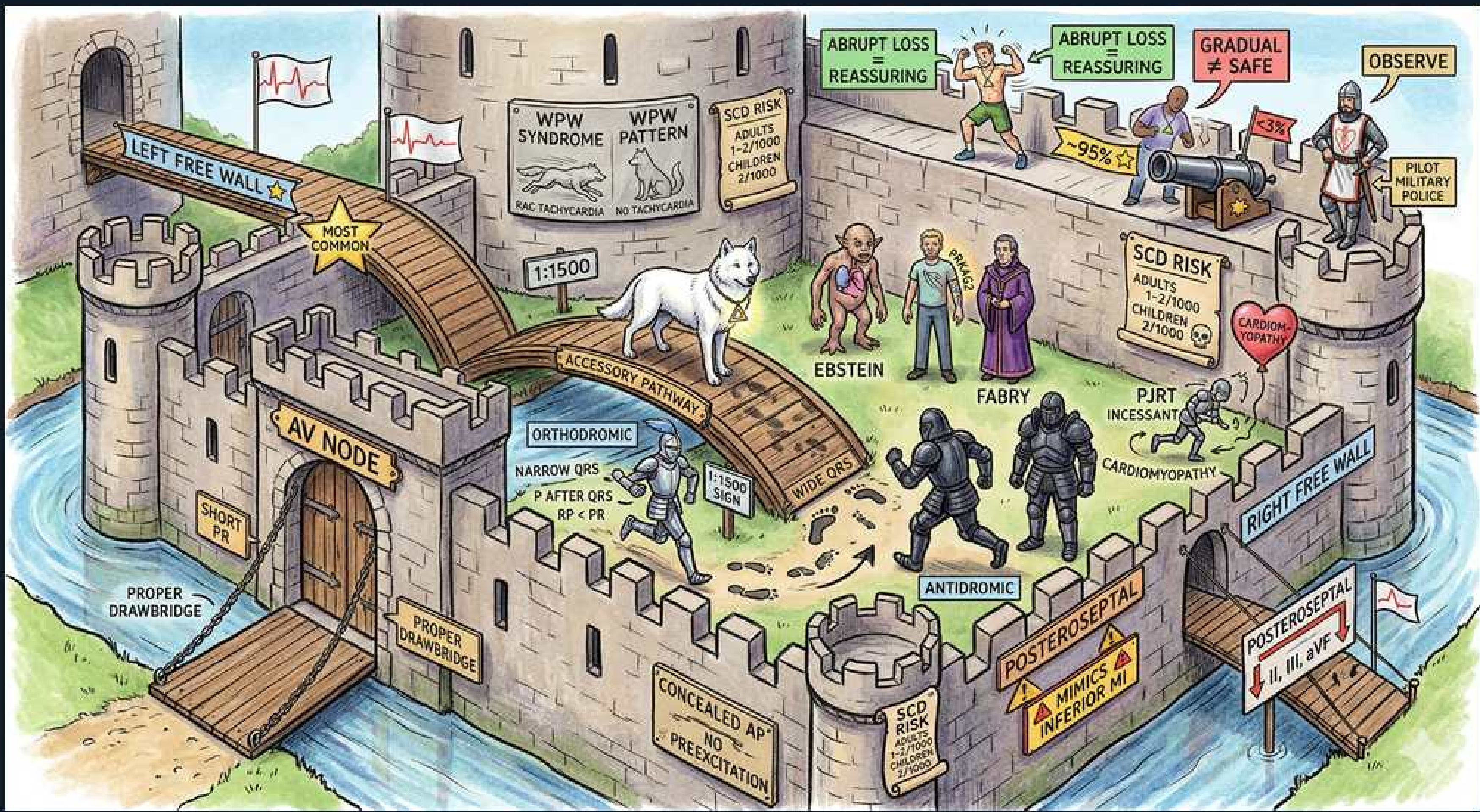
Arrhythmias — WPW & Accessory Pathways (AVRT)



1. AV node drawbridge

WHAT YOU SEE

AV NODE drawbridge gate, PROPER DRAWBRIDGE

WHAT IT ENCODES

The AV node is the normal, decremental gateway from atria to ventricles — it deliberately slows conduction (the PR interval) to protect the ventricles from fast atrial rates. Its decremental, rate-dependent slowing is exactly why AV-nodal blockers (adenosine, beta-blockers, calcium-channel blockers) terminate node-dependent re-entry. In WPW the danger is that an accessory pathway provides a SECOND route that bypasses this protective drawbridge.

BOARD TRAP

WRONG: 'The AV node is the only electrical connection between atria and ventricles, so blocking it always controls the ventricular rate.' The discriminator is the accessory pathway — in pre-excited AF, blocking the node shunts conduction down the bypass tract and can accelerate the ventricular rate to VF, so 'block the node' is NOT universally safe.

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2. Accessory pathway

WHAT YOU SEE

ACCESSORY PATHWAY side bridge

WHAT IT ENCODES

WPW features an accessory pathway (bundle of Kent) — an abnormal muscular bridge that connects atrium to ventricle outside the AV node, conducting rapidly and NON-decrementally. Because it pre-excites ventricular myocardium ahead of the AV-node impulse, the resting ECG shows a short PR (<120 ms) and a delta wave (slurred QRS upstroke) with a widened QRS. The same dual pathways (node + Kent) set up the re-entrant circuit that causes AVRT.

BOARD TRAP

WRONG: 'A short PR interval by itself establishes the diagnosis of WPW.' The discriminator is the delta wave — a short PR without a delta wave suggests LGL-type enhanced nodal conduction or a normal variant, whereas WPW requires the delta wave/pre-excitation from the accessory pathway.

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3. WPW syndrome vs pattern

WHAT YOU SEE

WPW SYNDROME and WPW PATTERN signs

WHAT IT ENCODES

WPW SYNDROME = ECG pre-excitation (delta wave + short PR) PLUS documented arrhythmia/symptoms (palpitations, SVT, syncope). WPW PATTERN = the same delta wave/short PR on ECG in an asymptomatic person with no arrhythmia. The distinction matters for management: symptomatic syndrome warrants electrophysiology study and ablation, while an asymptomatic pattern is risk-stratified (some high-risk asymptomatic patients, e.g., short accessory-pathway refractory period, are still offered ablation).

BOARD TRAP

WRONG: 'A delta wave on a routine ECG in an asymptomatic patient is WPW syndrome requiring immediate ablation.' The discriminator is symptoms/arrhythmia — without them this is WPW PATTERN, which is risk-stratified (often by EP study) rather than automatically ablated; calling it 'syndrome' overtreats.

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4. Orthodromic AVRT

WHAT YOU SEE

A white dog runs DOWN the AV-node drawbridge and back up the accessory bridge labeled 'ORTHODROMIC' (NARROW QRS, P after QRS) — looping forward through the node first gives the common narrow-complex circuit.

WHAT IT ENCODES

Orthodromic AVRT is the MOST common WPW tachycardia (~90–95%): the impulse goes DOWN the AV node and back UP the accessory pathway, so the ventricles are activated normally → NARROW QRS, regular, rate ~150–250, with a retrograde P wave AFTER the QRS (short RP). Because the AV node is part of the circuit, vagal maneuvers and AV-nodal blockers (adenosine 6 mg IV push, then 12 mg) terminate it — this is the one WPW arrhythmia where adenosine is appropriate.

BOARD TRAP

WRONG: 'A narrow-complex regular SVT in a known WPW patient is the same dangerous rhythm where adenosine is forbidden.' The discriminator is QRS width/regularity — narrow + regular = orthodromic AVRT (node-dependent, adenosine works); the contraindication to nodal blockers applies to the WIDE, IRREGULAR pre-excited AF, not to orthodromic AVRT.

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5. Most common direction

WHAT YOU SEE

MOST COMMON arrow

WHAT IT ENCODES

Orthodromic conduction (down the node, up the pathway) is the most common AVRT direction, accounting for roughly 90–95% of AVRT in WPW, and it produces a narrow-complex tachycardia. Antidromic AVRT (the wide-complex form) is far rarer.

BOARD TRAP

WRONG: 'The most common WPW tachycardia is a wide-complex rhythm, so it is hard to distinguish from VT.' The discriminator is prevalence and morphology — the COMMON form (orthodromic) is narrow-complex; only the uncommon antidromic form is wide and VT-mimicking.

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6. Antidromic AVRT

WHAT YOU SEE

ANTIDROMIC black knights, wide complex

WHAT IT ENCODES

Antidromic AVRT is the uncommon reverse circuit: DOWN the accessory pathway and UP the AV node, so the ventricles are activated entirely through the bypass tract → a maximally pre-excited WIDE-complex regular tachycardia that closely mimics VT. It is more frequent with multiple accessory pathways. Treatment leans toward procainamide/ibutilide or cardioversion rather than aggressive nodal blockade because the rhythm is pathway-dependent and may be hard to distinguish from VT.

BOARD TRAP

WRONG: 'A wide-complex regular tachycardia in a WPW patient should be treated as SVT with a calcium-channel blocker or adenosine.' The discriminator is that a wide-complex tachycardia is VT until proven otherwise AND, if it is antidromic AVRT, nodal blockers are risky — the safe move is procainamide or synchronized cardioversion, not verapamil/diltiazem.

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7. Concealed pathway

WHAT YOU SEE

CONCEALED AP, NO PRE-EXCITATION

WHAT IT ENCODES

A concealed accessory pathway conducts ONLY retrograde (ventricle→atrium), so there is NO delta wave or short PR on the resting ECG — the baseline tracing looks normal. It still completes the orthodromic re-entry loop and causes recurrent narrow-complex AVRT. Because it cannot conduct antegrade, a concealed pathway does NOT carry the pre-excited-AF/SCD risk of a manifest pathway.

BOARD TRAP

WRONG: 'A normal resting ECG rules out an accessory pathway as the cause of recurrent SVT.' The discriminator is retrograde-only conduction — a concealed pathway produces a normal baseline ECG yet still drives orthodromic AVRT, diagnosed at electrophysiology study, so a normal ECG does not exclude it.

8. Left free wall most common

WHAT YOU SEE

LEFT FREE WALL banner

WHAT IT ENCODES

The left free wall (left lateral) is the single most common accessory-pathway location, followed by posteroseptal, right free wall, and anteroseptal. Pathway location is inferred from delta-wave polarity across the 12 leads and confirmed at EP study, guiding the ablation target. Left free-wall pathways are typically ablated via a transseptal or retrograde-aortic approach.

BOARD TRAP

WRONG: 'Accessory pathways are most commonly located in the septum, so anteroseptal ablation is the default.' The discriminator is location frequency — the LEFT FREE WALL is most common; anteroseptal pathways are both less common and riskier to ablate (proximity to the AV node/His), so they are not the default target.

9. Ebstein anomaly

WHAT YOU SEE

A bluish, congenitally malformed figure labeled 'EBSTEIN' standing inside the castle — the deformed heart drawn here is the apically-displaced tricuspid valve that classically carries right-sided accessory pathways.

WHAT IT ENCODES

Ebstein anomaly (apically displaced, malformed tricuspid valve with an 'atrialized' RV) is the classic congenital association with WPW — these patients often have RIGHT-SIDED and frequently MULTIPLE accessory pathways. Suspect Ebstein when a young patient has pre-excitation plus tricuspid regurgitation, right-heart enlargement, and cyanosis/RBBB. Lithium exposure in utero is a known risk factor.

BOARD TRAP

WRONG: 'Multiple right-sided accessory pathways in a young patient point to an isolated electrical disease with a structurally normal heart.' The discriminator is the structural lesion — multiple right-sided pathways should prompt echo evaluation for Ebstein anomaly, which changes the procedural risk and overall management.

10. Fabry

WHAT YOU SEE

A figure labeled 'FABRY' beside Ebstein — represents the infiltrative storage disease whose glycolipid deposits shorten the PR and pre-excite the ventricle.

WHAT IT ENCODES

Fabry disease (X-linked alpha-galactosidase A deficiency with glycosphingolipid accumulation) is associated with a SHORT PR/pre-excitation and conduction abnormalities, alongside LVH, angiokeratomas, acroparesthesias, corneal verticillata, and renal failure. The infiltrative deposition shortens AV conduction early and later causes conduction disease/arrhythmias.

BOARD TRAP

WRONG: 'A short PR interval with LVH on ECG is simply athletic remodeling or hypertensive LVH.' The discriminator is the multisystem clues — a short PR plus LVH with neuropathic pain, angiokeratomas, and proteinuria should raise Fabry disease, which is treatable with enzyme-replacement therapy.

11. SCD risk 1:1000

WHAT YOU SEE

SCD RISK ~1:1000-2000 cannon

WHAT IT ENCODES

Manifest WPW carries a small but real annual sudden-cardiac-death risk (~1:1000 to 1:2000 per year), driven by pre-excited AF degenerating into VF when the accessory pathway conducts very rapidly to the ventricles. High-risk features include a SHORT accessory-pathway antegrade refractory period (<250 ms), shortest pre-excited RR in AF <250 ms, and multiple pathways. These features justify EP study and ablation even in some asymptomatic patients.

BOARD TRAP

WRONG: 'WPW is a benign ECG finding with negligible sudden-death risk, so no risk stratification is needed.' The discriminator is the accessory-pathway refractory period — a short refractory period allows lethally fast conduction in AF; such high-risk patients warrant EP testing and ablation rather than reassurance.

12. PJRT incessant

WHAT YOU SEE

PJRT INCESSANT, cardiomyopathy

WHAT IT ENCODES

Permanent junctional reciprocating tachycardia (PJRT) is a form of orthodromic AVRT using a SLOWLY conducting, decremental, concealed posteroseptal accessory pathway, producing a near-incessant long-RP narrow-complex tachycardia (inverted P in II/III/aVF). Because it runs almost continuously, it is a classic cause of tachycardia-induced cardiomyopathy in children/young adults; ablation is curative and reverses the cardiomyopathy.

BOARD TRAP

WRONG: 'A young patient with an incessant long-RP tachycardia and a dilated, poorly contracting LV has a primary dilated cardiomyopathy needing transplant evaluation.' The discriminator is the incessant tachycardia driving the dysfunction — recognizing tachycardia-induced cardiomyopathy from PJRT means ablation (not transplant) can reverse the LV dysfunction.

13. Septal localization

WHAT YOU SEE

POSTEROSEPTAL / ANTEROSEPTAL signs, II/III/aVF

WHAT IT ENCODES

Delta-wave polarity localizes the accessory pathway before ablation: a NEGATIVE delta wave in the inferior leads (II, III, aVF) suggests a POSTEROSEPTAL pathway, while pathway position across V1 and the frontal axis maps left- vs right-sided and septal locations. Anteroseptal/parahisian pathways sit close to the AV node and His bundle, raising the risk of inadvertent heart block during ablation.

BOARD TRAP

WRONG: 'Ablation risk is identical regardless of accessory-pathway location.' The discriminator is anatomy — anteroseptal/midseptal (parahisian) pathways carry a meaningfully higher risk of AV-block during ablation because of their proximity to the normal conduction system, so localization changes both approach and consent.